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SMART AUTO-FOLLOWING SHOPPING CART (W/ RFID)

CSE461
Spring 2026
Sec:03
Group: 06
Faculty: RDW, SMYA

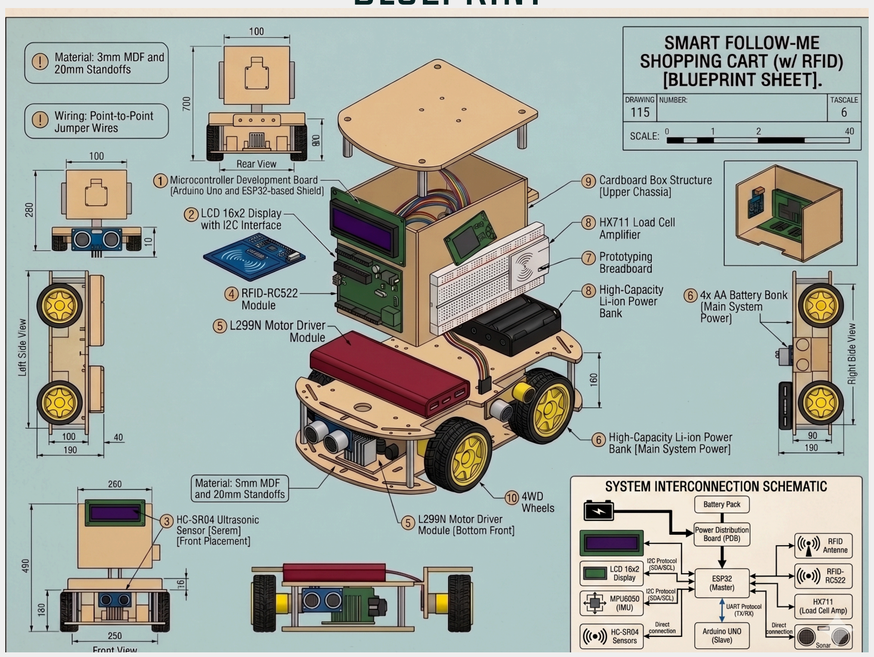
ABSTRACT

This project develops a self-directed Auto-Following Shopping Cart with an integrated RFID self-checkout system. Designed to ease retail experiences, it eliminates the burden of pulling carts and reduces checkout time. The system uses a dual-microcontroller setup (Arduino UNO R3 and ESP32), following users via Bluetooth RSSI, avoiding obstacles with ultrasonic sensors, and stabilizing movement through an MPU6050 IMU. Two RFID RC522 readers and HX711 load cells allow item scanning and real-time monitoring of cost and weight on an LCD display. The prototype demonstrates a low-cost, effective solution for small retail settings, enhancing accessibility for elderly and physically impaired shoppers while also serving as a simple budgeting tool for everyday users.

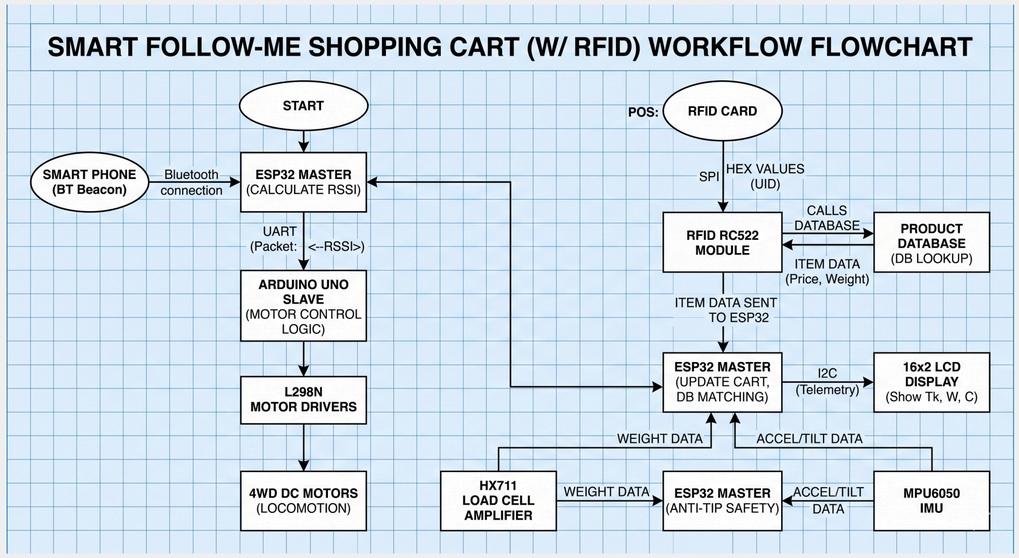
COMPONENTS LIST

- Arduino UNO R3
- ESP32 DevKit
- Smart Phone
- L298N Motor Driver
- DC Gear Motors (4WD) Chassis
- Ultrasonic Sensor HC-SR04
- MPU6050
- RFID Reader RC522
- RFID NFC Smart Card
- Load cell (10 kg) + HX711
- LCD 16x2 with I2C module
- Buttons
- Battery case 4xAA
- Battery 3.7V
- Power Distribution Board
- Jumper Wires (M-M, M-F, F-F)
- Resistors
- Breadboards (Small)

BLUEPRINT



FUNCTIONALITY WORKFLOW

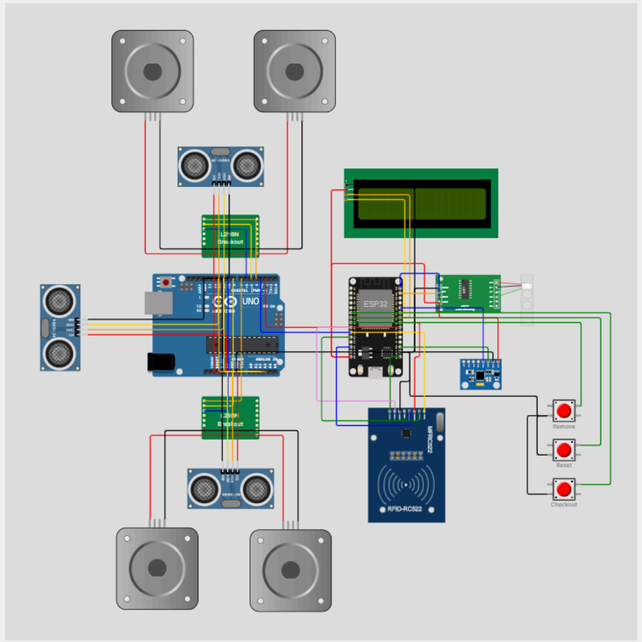


CHALLENGES

PROBLEMS	RESOLUTIONS
I2C PROTOCOL FAILURE — ESP32 & UNO SPEED MISMATCH PREVENTED STABLE I2C COMMUNICATION.E.	SWITCHED TO UART WITH FRAMED PACKETS (E.G. <-75>) AT 10HZ, WITH A 1000MS DEADMAN'S SWITCH TIMEOUT.
ULTRASONIC GHOST ECHOES — HC-SR04 RETURNED FALSE "0" READINGS MISREAD AS 999 CM, CAUSING ABRUPT STOPS.	ADDED A SOFTWARE FILTER REJECTING VALUES ≤ 0 CM AND ABOVE THE SAFE THRESHOLD.
4WD MOTOR STALLING — REAR WHEELS STALLED DUE TO POOR CURRENT DISTRIBUTION AND WRONG TB6612FNG PIN MAPPING.	UPGRADED TO DUAL L298N DRIVERS (ONE PER SIDE), REFACTORED TO HARDWARE PWM PINS D5/D9, SYNCHRONIZED IN PINS.

TOTAL COST: 4900 TAKA

CIRCUIT INTEGRATION



RESULTS/FINDINGS

TRACKING: THE USER FOLLOWING IN -50 TO -85 DBM RSSI IS RELIABLE AND THE SIGNAL NOISE IS FILTERED USING SOFTWARE.

OBSTACLE AVOIDANCE: INSTANT STOPS LESS THAN 30 CM, SIDE STEERING LESS THAN 20 CM, FALSE ECHOES DISREGARDED.

SAFETY: IMU LOCKED WHEELS 350 TILT DEADMAN SWITCH CUT MOTORS ON LOSS OF SIGNAL 1000MS.

RFID & POS: 100% SCAN ACCURACY. ANTI-THEFT IMPLEMENTATION WAS NOT POSSIBLE DUE TO LOAD CELL DRIFT. ALL BUTTONS FUNCTIONAL.

CONCLUSION

The Smart Follow-Me Shopping Cart is a highly cost-effective (4,900 BDT) autonomous prototype that uses IoT and sensor fusion to safely follow users, avoid obstacles, and process self-checkouts via RFID. Despite minor sensitivities to signal noise and vibration, it stands as a highly viable proof-of-concept with immense potential to eliminate retail queues and improve accessibility for disabled shoppers.